

9.7.1

Rushil Shanmukha Srinivas
EE25BTECH11057
Electrical Engineering ,
IIT Hyderabad.

October 11, 2025

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Problem Statement

Question : Find the area of the region between the conics given by $\{S(x, y) : x^2 + y^2 \leq 16a^2 \text{ and } y^2 \leq 6ax\}$.

Data

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - (6a \ 0) \mathbf{x} = 0$
Circle	$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 = 0$

Table: Parabola and Circle

Finding Area between Parabola and Circle

On comparing conics with

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (4.1)$$

The parameters of the Parabola are

$$\mathbf{V}_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_1 = -\begin{pmatrix} 3a \\ 0 \end{pmatrix}, f_1 = 0 \quad (4.2)$$

The parameters of the Circle are

$$\mathbf{V}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_2 = 0, f_2 = -16a^2 \quad (4.3)$$

The intersection of two conics with parameters $\mathbf{V}_i, \mathbf{u}_i, f_i$, $i = 1, 2$ is defined as

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0. \quad (4.4)$$

(4.1) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{vmatrix} = 0 \quad (4.5)$$

From (4.5) , (4.4) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (4.6)$$

On substituting from (4.2) and (4.3) in (4.6), we obtain

$$\begin{vmatrix} \mu & 0 & -3a \\ 0 & \mu + 1 & 0 \\ -3a & 0 & -16\mu a^2 \end{vmatrix} = 0 \quad (4.7)$$

On solving we get

$$(\mu + 1)(16\mu^2 + 9) = 0 \quad (4.8)$$

yielding

$$\mu = -1 \quad (4.9)$$

Substituting (4.2) and (4.3) in (4.4), we obtain

$$\mathbf{x}^\top \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2(-3a \ 0) \mathbf{x} + 16a^2 = 0 \quad (4.10)$$

On solving we get

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \text{ and } \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a \quad (4.11)$$

But circle and line $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a$ do not intersect. So The solution is

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \quad (4.12)$$

Therefore the intersection of two conics is a line with parameters

$$\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix} \quad (4.13)$$

The points of intersection for a conic and a line is given by

$$\kappa_i = \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \quad (4.14)$$

$$g(\mathbf{x}) = \mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 \quad (4.15)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h} - 16a^2 \quad (4.16)$$

The intersection parameters of the chord in (4.12) with the circle is obtained as

$$\kappa_i = \pm \sqrt{16a^2 - \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h}} \quad (4.17)$$

$$\kappa_i = \pm 2\sqrt{3}a \quad (4.18)$$

So the points of intersection are obtained as

$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \quad (4.19)$$

The desired area of region is given as

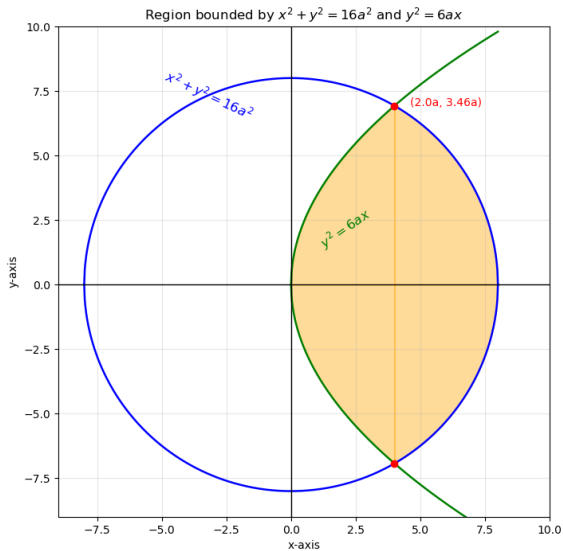
$$2 \left(\int_0^{2a} \sqrt{6ax} \, dx + \int_{2a}^{4a} \sqrt{16a^2 - x^2} \, dx \right) \quad (4.20)$$

$$= 2 \left[\sqrt{6a} \left(\frac{2}{3} x^{3/2} \right) \right]_0^{2a} + 2 \left[\frac{x}{2} \sqrt{16a^2 - x^2} + 8a^2 \sin^{-1} \left(\frac{x}{4a} \right) \right]_{2a}^{4a} \quad (4.21)$$

$$= \frac{4a^2}{\sqrt{3}} + \frac{16\pi a^2}{3} \quad (4.22)$$

$$= 4a^2 \left(\frac{4\pi}{3} + \frac{1}{\sqrt{3}} \right) \quad (4.23)$$

Plots



C Code

```
#include <stdio.h>
#include <math.h>
/*
    Program to find the area of the region:
     $S = \{ (x, y): x^2 + y^2 \leq 16a^2 \text{ and } y^2 \leq 6ax \}$ 

    We compute:
     $A = 2 * [ \int_0^{2a} \sqrt{6ax} \, dx + \int_{2a}^{4a} \sqrt{16a^2 - x^2} \, dx ]$ 
    using Simpson's 1/3 rule (numerical integration).
*/

// Function for the parabola  $y = \sqrt{6ax}$ 
double parabola(double x, double a) {
    return sqrt(6.0 * a * x);
}

// Function for the circle  $y = \sqrt{16a^2 - x^2}$ 
double circle(double x, double a) {
```

```

    return sqrt(16.0 * a * a - x * x);
}
// Simpsons rule for integration
double simpson(double (*f)(double, double), double a, double x0, double
x1, int n) {
    double h = (x1 - x0) / n;
    double sum = f(x0, a) + f(x1, a);
    for (int i = 1; i < n; i++) {
        double x = x0 + i * h;
        if (i % 2 == 0)
            sum += 2 * f(x, a);
        else
            sum += 4 * f(x, a);
    }
    return (h / 3.0) * sum;
}

```

```

int main() {
    double a;
    printf("Enter the value of a: ");
    scanf("%lf", &a);

    int n = 10000; // number of subintervals for Simpsons rule (even)

    // Integrate sqrt(6ax) from 0 to 2a
    double area1 = simpson(parabola, a, 0, 2 * a, n);

    // Integrate sqrt(16a^2 - x^2) from 2a to 4a
    double area2 = simpson(circle, a, 2 * a, 4 * a, n);

    // Total area (both above and below x-axis)
    double total_area = 2 * (area1 + area2);

    // Display results
    printf("\n--- Step-by-Step Integration Process ---\n");
}

```

```

printf(" Parabola:  $y = \sqrt{6ax}$ \n");
printf(" Circle:  $y = \sqrt{16a^2 - x^2}$ \n");
    printf(" Intersection point: (2a, 23 a)\n");

    printf("\nIntegrating in first quadrant:\n");
    printf(" Area1 = [0 to 2a]  $\sqrt{6ax}$  dx = %.6lf\n", area1);
    printf(" Area2 = [2a to 4a]  $\sqrt{16a^2 - x^2}$  dx = %.6lf\n", area2);

    printf("\nTotal Area (both halves):  $A = 2*(Area1 + Area2)$ \n");
    printf("=> Area = %.6lf\n", total_area);

    return 0;
}

```

```
import ctypes
import os
# Load the compiled shared object (.so file)
# Make sure "area_region.so" is in the same directory
so_file = ". /area_region.so"

# Load the shared object
lib = ctypes.CDLL(so_file)

# Define the argument and return types for the C function 'main'
lib.main.restype = ctypes.c_int

print("Running the C integration program... \n")
# Call the main() function from the C program
lib.main()
```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt
# Parameter 'a'
a = 2 # You can change this value

# Circle:  $x^2 + y^2 = 16a^2$   $y = (16a^2 - x^2)$ 
x_circle = np.linspace(-4 * a, 4 * a, 800)
y_circle_upper = np.sqrt(16 * a**2 - x_circle**2)
y_circle_lower = -y_circle_upper

# Parabola:  $y^2 = 6ax$   $y = (6ax)$ 
x_parabola = np.linspace(0, 4 * a, 500)
y_parabola_upper = np.sqrt(6 * a * x_parabola)
y_parabola_lower = -y_parabola_upper

# Intersection points (analytical)
x_intersect = 2 * a
```



```
y_intersect = 2 * np.sqrt(3) * a
# Create plot
plt.figure(figsize=(8, 8))

# Shaded region between curves (upper and lower parts)
x_fill1 = np.linspace(0, x_intersect, 300)
x_fill2 = np.linspace(x_intersect, 4 * a, 300)
y_fill1_upper = np.sqrt(6 * a * x_fill1)
y_fill2_upper = np.sqrt(16 * a**2 - x_fill2**2)

plt.fill_between(x_fill1, y_fill1_upper, color='orange', alpha=0.4)
plt.fill_between(x_fill2, y_fill2_upper, color='orange', alpha=0.4)
plt.fill_between(x_fill1, -y_fill1_upper, color='orange', alpha=0.4)
plt.fill_between(x_fill2, -y_fill2_upper, color='orange', alpha=0.4)

# Plot circle
plt.plot(x_circle, y_circle_upper, 'b', linewidth=1.8)
plt.plot(x_circle, y_circle_lower, 'b', linewidth=1.8)
```

```

# Plot parabola
plt.plot(x_parabola, y_parabola_upper, 'g', linewidth=1.8)
plt.plot(x_parabola, y_parabola_lower, 'g', linewidth=1.8)

# Mark intersection points
plt.scatter([x_intersect, x_intersect], [y_intersect, -y_intersect], color='red',
            zorder=5)
plt.text(x_intersect + 0.3 * a, y_intersect, f'({x_intersect/a:.1f}a, {
            y_intersect/a:.2f}a)',
            color='red', fontsize=10)

plt.text(-2.5 * a, 3.2 * a, r'$x^2 + y^2 = 16a^2$', color='blue', fontsize
        =12, rotation=-25)
plt.text(0.5 * a, 0.7 * a, r'$y^2 = 6ax$', color='green', fontsize=12,
        rotation=35)

# Axes setup
plt.axhline(0, color='black', linewidth=1)

```

```
plt.axvline(0, color='black', linewidth=1)
plt.gca().set_aspect('equal', adjustable='box')
plt.xlim(-4.5 * a, 5.0 * a)
plt.ylim(-4.5 * a, 5.0 * a)

# Title and labels
plt.title(r"Region bounded by  $x^2 + y^2 = 16a^2$  and  $y^2 = 6ax$ ")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.grid(alpha=0.3)
plt.savefig("../figs/fig16.png")
# Show plot
plt.show()
```